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Negative-Mass Antimatter and the Baryon Asymmetry Problem

A CPT-Symmetric Cosmological Framework

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ABSTRACT

The observed baryon asymmetry of the universe, the overwhelming dominance of matter over antimatter, remains one of the deepest unsolved problems in modern physics. The Standard Model meets the Sakharov conditions for baryogenesis only in principle: its CP violation is far too weak, and its electroweak transition too smooth, to produce the observed asymmetry ratio of $\eta_B \approx 6.1 \times 10^{-10}$. This report examines an alternative hypothesis: that antimatter possesses negative gravitational mass, joined to the conjecture that matter and antimatter were segregated in time at the Big Bang into CPT-conjugate sectors, so that the universe as a whole holds equal amounts of each. We show that the

Feynman–Stückelberg interpretation of antiparticles does not support that conjecture; it reinterprets a negative-energy solution running backward in time as a positive-energy antiparticle running forward in time, so real antimatter has positive energy. We review the relevant physics: Bondi's (1957) negative mass framework in general relativity, the Feynman–Stückelberg interpretation, Villata's (2011) CPT-based argument for matter–antimatter gravitational repulsion, the Dirac-Milne universe of Benoit-Lévy and Chardin (2012), and Farnes' (2018) negative-mass cosmology, and we treat the Boyle–Finn–Turok (2018) CPT-symmetric universe as a separate idea. We examine the implications for Big Bang nucleosynthesis, the CMB, dark energy, and structure formation, and evaluate the constraints from the ALPHA-g (2023), AEgIS, and GBAR experiments at CERN. A constant negative-mass density acts as a negative cosmological constant, not a positive one, and Farnes' negative masses, which obey the equivalence principle, cannot be the antimatter this hypothesis describes. The simplest version of the hypothesis (antigravity for antihydrogen) is ruled out by ALPHA-g, which measured antihydrogen falling toward Earth at $(0.75 \pm 0.13 \pm 0.16)\ g$. The more sophisticated versions concede that laboratory antimatter gravitates normally, which leaves no repulsion to drive segregation, and they make no quantitative predictions. We conclude that the negative-gravitational-mass route to the baryon asymmetry is effectively excluded. The CPT-symmetric universe of Boyle, Finn, and Turok, which uses no negative mass and explains the matter excess by thermal leptogenesis, remains a distinct and testable research program.

KEYWORDS: BARYON ASYMMETRY, ANTIMATTER, NEGATIVE MASS, CPT SYMMETRY, ANTIGRAVITY, DARK ENERGY, FEYNMAN–STÜCKELBERG INTERPRETATION, COSMOLOGICAL CONSTANT

A Speculative Theoretical Investigation

SECTION 1

1. Introduction

1.1 The Matter–Antimatter Asymmetry Problem

One of the most profound puzzles in contemporary physics is deceptively simple to state: Why is there something rather than nothing? More precisely, why does the observable universe consist almost entirely of matter, with antimatter appearing only in trace quantities produced by high-energy processes such as cosmic ray interactions and radioactive decay?

The Standard Model of particle physics, combined with the hot Big Bang cosmological model, predicts that the early universe should have produced matter and antimatter in nearly equal quantities. At temperatures above the QCD confinement scale ($T \gtrsim 200$ MeV), quarks and antiquarks existed in thermal equilibrium, continuously created and annihilated in pairs. As the universe cooled, these pairs should have mutually annihilated, leaving behind a universe of pure radiation — photons and neutrinos — with essentially no residual baryonic matter.

Yet we observe precisely the opposite: a universe filled with galaxies, stars, planets, and observers, composed overwhelmingly of matter. The baryon-to-photon ratio, as measured by the cosmic microwave background (CMB) anisotropies by the Planck satellite, is (Planck Collaboration, 2020; Navas et al., 2024):

$$\eta_B = (n_B - n_{\bar{B}}) / n_\gamma = (6.12 \pm 0.04) \times 10^{-10}$$

This number, while small, represents an enormous asymmetry: for every billion antimatter particles produced in the Big Bang, there were approximately one billion and one matter particles. The billion pairs annihilated, and the lone excess matter particle survived to form everything we observe.

The absence of significant antimatter in the observable universe is confirmed by multiple independent lines of evidence:

1. Cosmic ray composition: The observed flux of antiprotons in cosmic rays is consistent with secondary production from high-energy collisions and shows no evidence of primary antimatter sources (AMS-02 Collaboration, Aguilar et al., 2016).

1. Gamma-ray background: If regions of antimatter existed adjacent to regions of matter, their boundaries would produce annihilation radiation at 511 keV and at the pion mass scale (~135 MeV). No such signatures have been observed at cosmological scales (Cohen, De Rújula, & Glashow, 1998).

1. Big Bang nucleosynthesis (BBN): The observed abundances of light elements (^{4}He, D, ^{3}He, ^{7}Li) are consistent with a matter-dominated universe at $T \sim 1$ MeV and provide independent confirmation of η_B (Fields, Olive, Yeh, & Young, 2020).

1. CMB anisotropies: The acoustic peak structure of the CMB power spectrum is sensitive to the baryon-to-photon ratio and yields values fully consistent with BBN (Planck Collaboration, 2020).

1.2 The Sakharov Conditions and Their Insufficiency

In a landmark 1967 paper, Andrei Sakharov identified three necessary conditions for the dynamical generation of a baryon asymmetry from an initially symmetric state (Sakharov, 1967):

1. Baryon number (B) violation: Interactions must exist that change the net baryon number.

1. C and CP violation: Charge conjugation symmetry and the combined charge-parity symmetry must be violated, so that processes creating baryons proceed at different rates than those creating antibaryons.

1. Departure from thermal equilibrium: The baryon-number-violating, CP-violating processes must occur out of thermal equilibrium, since CPT symmetry guarantees that in equilibrium, any asymmetry-generating process is exactly compensated by its CPT conjugate.

The Standard Model, remarkably, satisfies all three conditions in principle:

- B violation occurs through non-perturbative electroweak sphaleron processes, which violate B + L while conserving B - L ('t Hooft, 1976; Klinkhamer & Manton, 1984).

- CP violation is present in the CKM quark mixing matrix, as demonstrated experimentally in the kaon and B-meson systems (Christenson, Cronin, Fitch, & Turlay, 1964; BaBar Collaboration, 2001; Belle Collaboration, 2001).

- Departure from equilibrium could occur during the electroweak phase transition if it is first-order.

However, the Standard Model fails quantitatively on at least two fronts:

First, the amount of CP violation in the CKM matrix is insufficient by many orders of magnitude. The Jarlskog invariant $J \approx 3 \times 10^{-5}$, and when combined with the relevant mass scales, produces an asymmetry of order $\eta_B^{\text{SM}} \sim 10^{-18}$, roughly eight orders of magnitude too small (Gavela, Hernández, Orloff, & Pène, 1994; Huet & Sather, 1995).

Second, lattice gauge theory calculations have demonstrated that the electroweak phase transition in the Standard Model (with the observed Higgs mass of $m_H \approx 125$ GeV) is a smooth crossover rather than

a first-order phase transition, invalidating the third Sakharov condition within the Standard Model framework (Kajantie, Laine, Rummukainen, & Shaposhnikov, 1996).

These failures have motivated an enormous body of work on beyond-Standard-Model (BSM) baryogenesis mechanisms, including:

- GUT baryogenesis via the decay of superheavy bosons (Yoshimura, 1978; Weinberg, 1979)
- Electroweak baryogenesis with extended Higgs sectors providing a first-order phase transition and additional CP violation (Morrissey & Ramsey-Musolf, 2012)
- Leptogenesis via the out-of-equilibrium decay of heavy right-handed neutrinos, with subsequent sphaleron conversion of lepton asymmetry to baryon asymmetry (Fukugita & Yanagida, 1986)
- Affleck-Dine baryogenesis using flat directions in supersymmetric theories (Affleck & Dine, 1985)

Despite decades of effort, none of these mechanisms has been experimentally confirmed, and the baryon asymmetry problem remains open.

1.3 An Alternative Perspective: No Asymmetry At All?

This report explores a radically different approach. Rather than seeking a dynamical mechanism to generate a matter–antimatter asymmetry, we ask: What if no asymmetry was ever generated? What if the Big Bang produced exactly equal quantities of matter and antimatter, and the apparent absence of antimatter is an observational artifact arising from a fundamental property of antimatter that we have not yet correctly identified?

Specifically, we investigate the hypothesis that:

1. Antimatter possesses negative gravitational mass, causing it to be gravitationally repelled by matter.

1. Under the Feynman–Stückelberg interpretation, antiparticles can be described as particles propagating backward in time. As §3.5 shows, the interpretation in fact assigns real antiparticles positive energy, so it gives no support to item 1.

1. At the Big Bang, the CPT symmetry of the laws of physics was exactly realized: equal matter and antimatter were created, and the antimatter ended up in the opposite temporal direction, forming a CPT-mirror image of our universe. This step is a conjecture; nothing in this report derives it (§3.1, §3.4).

1. The observed baryon "asymmetry" is therefore not an asymmetry of the universe as a whole, but merely a feature of our local temporal sector.

This hypothesis, while speculative, draws on threads in theoretical physics developed separately by Villata (2011), Benoit-Lévy and Chardin (2012), Boyle, Finn, and Turok (2018), and Farnes (2018), though none of them proposes the combination examined here. Its most direct prediction, that antihydrogen falls upward, has now been tested at CERN, and the result went against it (Anderson et al., 2023).

1.4 Scope and Organization

This report is structured as follows. Section 2 establishes the theoretical framework, reviewing the Feynman–Stückelberg interpretation of antiparticles, CPT symmetry, and the physics of negative mass in general relativity and quantum field theory. Section 3 develops the negative mass–antimatter hypothesis formally, connecting it to CPT-symmetric cosmological models. Section 4 explores the cosmological implications for nucleosynthesis, the CMB, dark energy, and structure formation. Section 5 critically evaluates the experimental evidence, particularly the recent ALPHA-g results from CERN. Section 6 examines the relationship between negative-mass antimatter and the dark sector (dark matter and dark energy). Section 7 presents the mathematical formalism in detail. Section 8 addresses

criticisms and counterarguments with intellectual honesty. Section 9 offers conclusions and identifies future directions for research.

SECTION II

2. Theoretical Framework

2.1 The Feynman–Stückelberg Interpretation

The idea that antiparticles can be interpreted as particles traveling backward in time originated independently with Ernst Stückelberg (1941, 1942) and Richard Feynman (1949). In Feynman's formulation of quantum electrodynamics, a positron is mathematically equivalent to an electron propagating backward in time:

$$\psi_{\text{positron}}(\mathbf{x}, t) \sim \psi^*_{\text{electron}}(\mathbf{x}, -t)$$

More precisely, in the Feynman propagator formalism, the negative-energy solutions of the Dirac equation that propagate backward in time are reinterpreted as positive-energy antiparticles propagating forward in time. This is not merely a mathematical trick; it is built into the structure of quantum field theory. The Feynman propagator for a scalar field is:

$$\Delta_F(x - y) = \int [\mathrm{d}^4 p \,/\, (2\pi)^4] \cdot [i\, e^{-ip\cdot(x\,-\,y)} \,/\, (p^2 - m^2 + i\varepsilon)]$$

The iε prescription ensures that positive-energy solutions propagate forward in time while negative-energy solutions propagate backward in time. Under charge conjugation, the backward-propagating negative-energy electron becomes a forward-propagating positive-energy positron.

In Feynman diagrams, this interpretation is manifest: an electron line with an arrow pointing backward in time is identified with a positron line pointing forward in time. This is a fundamental feature of relativistic quantum field theory and has been experimentally validated to extraordinary precision through the success of QED calculations.

The key question is whether this mathematical equivalence between "antiparticle going forward in time" and "particle going backward in time" has deeper physical significance, or whether it is merely a convenient computational device. The hypothesis under investigation takes the former view seriously.

2.2 CPT Symmetry

The CPT theorem, proved by Lüders (1954), Pauli (1955), and Jost (1957), states that any Lorentz-invariant local quantum field theory with a Hermitian Hamiltonian is invariant under the combined operation of:

- C (charge conjugation): replacing particles with antiparticles
- P (parity): spatial inversion x → -x
- T (time reversal): t → -t

Formally, for any quantum field theory satisfying the Wightman axioms:

$$\Theta \, \mathcal{L}(x) \, \Theta^{-1} = \mathcal{L}(-x)$$

where $\Theta = \mathrm{CPT}$ is the combined CPT operator. This means that the CPT-transformed version of any physical process has exactly the same probability amplitude as the original process. Equivalently:

$$\mathcal{M}(\mathrm{A} \to \mathrm{B}) = \mathcal{M}(\overline{\mathrm{B}} \to \overline{\mathrm{A}})$$

where $\overline{\mathrm{A}}$ and $\overline{\mathrm{B}}$ are the CPT images of the initial and final states: every particle replaced by its antiparticle, with momenta unchanged and spins reversed.

CPT symmetry has been tested to extraordinary precision. Among the most stringent tests, the masses of the neutral kaon and its antiparticle agree to better than 6 parts in 10^{19} (Navas et al., 2024), and the magnetic g factors of the electron and positron agree to about two parts in 10^{12} (Van Dyck, Schwinberg, & Dehmelt, 1987).

The relevance of CPT to our hypothesis is profound. Under CPT:

- A particle with positive mass and positive charge maps to an antiparticle with the same positive inertial mass and the opposite charge, taking part in the time-reversed, mirror-image process.

The question is whether CPT merely relates the internal quantum numbers of particles and antiparticles (as is the standard interpretation) or whether it also relates their gravitational properties (as proposed by Villata and others).

2.3 Negative Mass in General Relativity

The concept of negative mass has a long and respectable history in general relativity. Hermann Bondi (1957) provided the first systematic analysis, distinguishing between three types of mass:

1. Inertial mass (m_i): the resistance to acceleration, appearing in $F = m_\mathrm{i} a$.

1. Active gravitational mass (m_a): the source of gravitational field.

1. Passive gravitational mass (m_p): the response to an external gravitational field.

The equivalence principle asserts that $m_i = m_p$. It has been tested to extraordinary precision: the MICROSCOPE satellite bounded the Eötvös parameter for titanium and platinum, the fractional difference in their free-fall accelerations, at about 1.3×10^{-14} in its first results (Touboul et al., 2017) and at the 10^{-15} level in its final analysis (Touboul et al., 2022). Conservation of momentum requires $m_a = m_p$. Thus, in standard physics, all three masses are equal.

Bondi showed that if all three masses are simultaneously negative (maintaining the equivalence principle), the dynamics are self-consistent but exotic. For two point masses M > 0 and m < 0 interacting gravitationally:

- The positive mass M attracts both positive and negative masses.
- The negative mass m repels both positive and negative masses.
- The negative mass accelerates *toward* the positive mass (because $F = m_i a$ with $m_i < 0$ reverses the direction of acceleration).
- The positive mass accelerates *away from* the negative mass.

This produces the famous runaway pair: the negative mass chases the positive mass, both accelerating indefinitely without violating conservation of energy or momentum (since the kinetic energy of the negative-mass object is itself negative). As Bondi noted, this is peculiar but not inconsistent.

Bonnor (1989) extended Bondi's analysis and examined the behavior of negative mass in various general-relativistic contexts, including the Schwarzschild solution and cosmological models. He showed

that a negative-mass Schwarzschild solution is repulsive and produces a naked singularity rather than a black hole.

Forward (1990) explored potential technological applications of negative mass, including propulsion systems. While highly speculative, Forward's work helped maintain theoretical interest in the concept.

2.4 Negative Mass in Quantum Field Theory

In quantum field theory, the concept of negative mass is more constrained but not entirely excluded. The mass parameter m in the Lagrangian density:

$$\mathcal{L} = \overline{\psi}(i\gamma^{\mu}\partial_{\mu} - m)\psi$$

is conventionally taken to be positive. However, this is a convention related to the choice of vacuum state rather than a fundamental requirement. The physical mass is the pole of the propagator:

$$S_F(p) = i(\gamma^{\mu}p_{\mu} + m) / (p^2 - m^2 + i\varepsilon)$$

and what is physically measurable is m^2, not the sign of m itself. In the standard interpretation, both signs of m yield the same physics for free fields.

The situation changes when gravity is included. In the semiclassical approximation, the source of gravity is the stress-energy tensor, written here in the (+, −, −, −) signature with $\hbar = c = 1$, as in the rest of this subsection:

$$T_{\mu\nu} = (2 / \sqrt{-g})\, \delta S_{\text{matter}} / \delta g^{\mu\nu}$$

For a Dirac field, the energy density (which serves as the source of gravity in the Newtonian limit) is, summing over the spatial index j:

$$T^{00} = \overline{\psi}(-i\gamma^j \partial_j + m)\psi$$

The sign of the mass term contributes to the gravitational source with a definite sign. This opens the theoretical possibility that if the mass parameter for antimatter fields were effectively negative in its gravitational coupling, antimatter would gravitate differently than matter. After quantization, however, particle and antiparticle excitations carry the same positive energy (§2.5), and the sign of m can be removed by a field redefinition (§7.5), so this possibility requires a nonstandard coupling between antimatter and gravity.

2.5 The Dirac Sea Reinterpretation

Dirac's original (1930) interpretation of the negative-energy solutions of his equation involved a "sea" of filled negative-energy states. A hole in this sea — an absence of a negative-energy, negative-charge electron — manifests as a positive-energy, positive-charge particle: the positron.

This picture, while superseded by the modern QFT formalism, offers an interesting perspective on the mass of antimatter. In the Dirac sea picture, a positron is the absence of a negative-energy electron. If we assign gravitational properties based on the energy of the state, the absence of negative energy is equivalent to the presence of positive energy — and the positron should gravitate normally.

However, this argument can be turned around. If we consider the negative-energy electron as having negative gravitational mass, then its absence (the positron) has the opposite gravitational mass — namely, positive gravitational mass in its rest frame. But this depends crucially on whether the "negative energy" of the Dirac sea has gravitational reality.

In the modern QFT reinterpretation, the vacuum is the state of minimum energy, and all particles (including antiparticles) have positive energy by construction. The gravitational properties of antiparticles are then determined by the positive-energy stress-energy tensor they carry. This standard argument predicts that antimatter falls down, not up.

The tension between the Dirac sea picture (where antiparticles are "holes" in a negative-energy sea) and the modern QFT picture (where antiparticles are positive-energy excitations) remains a source of conceptual ambiguity regarding the gravitational properties of antimatter — an ambiguity that can ultimately only be resolved experimentally.

2.6 Previous Theoretical Work on Antimatter Gravity

The gravitational behavior of antimatter has been debated since the 1950s. The early theoretical arguments, by Morrison, Schiff, and Good, went against repulsive gravity for antimatter. Later authors examined the loopholes in those arguments, and a smaller group has argued for repulsion:

Morrison (1958) gave an energy-conservation argument against antigravity. If matter and antimatter responded oppositely to gravity, lifting a particle–antiparticle pair would cost no work, yet the photons from annihilating the pair at the top would gain energy as they fell back down. A cycle of pair creation, lifting, and annihilation would then produce energy from nothing.

Schiff (1958, 1959) argued from the virtual electron–positron pairs that contribute to the mass of ordinary atoms in amounts that differ from element to element. A negative gravitational mass for the positron would then have appeared as a composition-dependent violation of the equivalence principle

in the Eötvös experiments, and none was seen, so he concluded that the positron's gravitational mass is positive. Nieto and Goldman (1991) regarded the renormalization steps in this argument as inconclusive.

Good (1961) argued from the neutral kaon system. If the $\overline{\mathrm{K}}^0$ fell up while the K^0 fell down, the difference in their gravitational energies would regenerate short-lived kaons in a beam of long-lived ones and produce two-pion decays at an unacceptably high rate. Such decays were later observed and attributed to CP violation (Christenson, Cronin, Fitch, & Turlay, 1964), and the argument has been criticized for relying on the absolute value of the gravitational potential.

Nieto and Goldman (1991) reviewed these arguments and concluded that each rules out some models of antigravity but not all, so that a direct measurement of the gravitational acceleration of antimatter remained worthwhile.

Chardin and Rax (1992) proposed that CP violation in the kaon system could be explained as a gravitational effect if antimatter experienced repulsive gravity.

Villata (2011) applied CPT to the classical equations of general relativity, including the geodesic equation:

$$\mathrm{d}^2x^\mu/\mathrm{d}\tau^2 + \Gamma^\mu{}_{\alpha\beta}\,(\mathrm{d}x^\alpha/\mathrm{d}\tau)(\mathrm{d}x^\beta/\mathrm{d}\tau) = 0$$

He noted that the acceleration term and the Christoffel symbols are both odd under the combined PT inversion, as is the four-momentum, which he treats as the gravitational "charge." Transforming an antimatter test body moving in the field of a matter source, but not the source itself, reverses the sign of its acceleration. He concluded that matter and antimatter repel each other gravitationally while each is

self-attractive, with all masses remaining positive. Section 3.2 discusses the argument and a published critique of it.

Benoit-Lévy and Chardin (2012) developed a cosmological model (the "Dirac-Milne" universe) based on matter–antimatter gravitational repulsion and argued that it could account for several cosmological observations without dark energy or inflation; §4.1 reviews its difficulties with nucleosynthesis.

SECTION III

3. The Negative Mass–Antimatter Hypothesis

3.1 Statement of the Hypothesis

We now develop the central hypothesis of this report in formal terms. The hypothesis consists of four interconnected claims, the third of which is a conjecture rather than a result:

Claim 1 (Negative Gravitational Mass): Antimatter possesses negative gravitational mass. Both the active gravitational mass (which sources the field) and the passive gravitational mass (which responds to it) of an antiparticle have the opposite sign to those of the corresponding particle:

$$m_{\mathrm{g}}(\overline{\mathrm{p}}) = -m_{\mathrm{g}}(\mathrm{p})$$

while the inertial mass stays positive and equal to that of the particle, as CPT requires (ensuring positive kinetic energy in the local frame):

$$m_i(\bar{p}) = m_i(p) > 0$$

This means the equivalence principle is violated for antimatter, with $m_g/m_i = -1$ for antiparticles. The passive mass must flip for antimatter to fall up in Earth's field. Flipping only the active mass would leave antihydrogen falling down while it pushed Earth away, which breaks the equality of active and passive mass that momentum conservation requires (Bondi, 1957).

Claim 2 (Gravitational Repulsion): As a consequence of Claim 1, matter and antimatter gravitationally repel each other:

$$\mathbf{F}_{grav}(p, \bar{p}) = +G\,|m_g(p)|\,|m_g(\bar{p})|\,/\,r^2\,\hat{\mathbf{r}} \quad \text{(repulsive)}$$

while antimatter attracts antimatter, because the product of two negative gravitational masses is positive. In this scheme like gravitational "charges" attract and unlike charges repel, the reverse of electromagnetism.

Conjecture 3 (Temporal Segregation): At the Big Bang singularity (or more precisely, at the CPT-symmetric boundary), antimatter ended up in the $t < 0$ direction while matter evolved into $t > 0$. This is an unmotivated conjecture. Nothing in Claims 1 and 2 implies it: gravitational repulsion is a force acting in space, and no equation of motion moves matter or antimatter into a time direction. Nor does the Feynman–Stückelberg interpretation support it. That interpretation reads a negative-energy solution running backward in time as a positive-energy antiparticle running forward in time (§2.1), so real antimatter has positive energy and, under the standard coupling of gravity to energy, gravitates normally (§2.5). Section 3.5 shows why the attempted link fails. We keep the conjecture only to explore its consequences, and we flag each result below that depends on it.

Claim 4 (CPT-Symmetric Cosmology): The universe as a whole is exactly CPT-symmetric. What we observe as the "universe" is only the t > 0 matter-dominated half. The t < 0 half is an antimatter-dominated, parity-reversed, time-reversed mirror image. The baryon asymmetry is not an asymmetry of physics but an anthropic selection effect: observers made of matter can only exist in the matter-dominated sector.

Claim 4 does not need Claims 1 to 3. A CPT-symmetric universe of this kind has been built without any negative mass by Boyle, Finn, and Turok (2018), discussed in §3.3. In their model the matter excess in our half is generated by leptogenesis, with the opposite excess in the mirror half, rather than being a selection effect alone.

3.2 Connection to CPT Symmetry and General Relativity

Villata (2011) gave a CPT-based argument for antigravity, which Cabbolet (2012) has criticized. In Villata's version all masses stay positive; the sign change comes from the behavior of the test body's trajectory under the PT inversion. A simplified Newtonian version of the argument proceeds as follows.

In general relativity, the motion of a test particle in a gravitational field is described by the geodesic equation. Consider an antimatter test particle moving in the field of a matter source.

The geodesic equation in the weak-field limit gives:

$$\mathrm{d}^2\mathbf{r}/\mathrm{d}t^2 = -\nabla\Phi$$

where $\Phi = -GM/r$ is the Newtonian gravitational potential of the source mass M.

Following Villata, apply the PT inversion ($x^\mu \to -x^\mu$) to the antimatter test particle's trajectory but not to the field of the matter source. Under T, $t \to -t$ leaves $d^2\mathbf{r}/dt^2$ unchanged (the double time derivative is T-even). Under P, $\mathbf{r} \to -\mathbf{r}$, so $d^2\mathbf{r}/dt^2 \to -d^2\mathbf{r}/dt^2$. The source's field, untransformed, keeps its gradient $\nabla\Phi$. The transformed equation becomes:

$$-d^2\mathbf{r}/dt^2 = -\nabla\Phi$$

which gives:

$$d^2\mathbf{r}/dt^2 = +\nabla\Phi$$

This is a repulsive gravitational interaction. The antiparticle is repelled by the matter source. If the source's field is transformed as well, its gradient also changes sign, the two signs cancel, and antimatter falls normally toward antimatter.

The critical and controversial step is applying the transformation to the antimatter body but not to the field of the matter source. In standard general relativity, mass-energy is positive, C does not affect it, and every test body follows the same geodesics whatever its composition. Villata's argument therefore requires an extension of CPT to the gravitational sector in a specific way that is not mandated by the CPT theorem. That theorem has been extended to quantum fields on curved spacetimes (Hollands, 2004), but the extension concerns quantum fields on a fixed background geometry and does not say that antimatter's gravitational mass changes sign. Cabbolet (2012) objected that Villata assumes the CPT symmetry he should be deriving from a theory of gravitational processes, and that the result conflicts with general relativity, in which a neutron and an antineutron with identical initial conditions in the same gravitational field follow the same trajectory.

3.3 The Boyle–Finn–Turok CPT-Symmetric Universe

In 2018, Boyle, Finn, and Turok proposed a cosmological model that realizes Claim 4 without negative mass of any kind (Boyle, Finn, & Turok, 2018). It is a separate idea from the negative-mass hypothesis and should be judged on its own terms. They noted that the Standard Model of particle physics, combined with general relativity, is CPT symmetric. They then asked: What if the universe itself respects this symmetry?

Their proposal is that the universe does not begin at the Big Bang but rather extends through it, with the Big Bang serving as a fixed point of a CPT transformation. Before the Big Bang (in the $t < 0$ direction), the universe is the CPT mirror of the $t > 0$ universe:

- Matter is replaced by antimatter (C)
- Left and right are reversed (P)
- Time runs in the opposite direction (T)

In this picture, the universe is a CPT-symmetric pair, analogous to a particle–antiparticle pair created from the vacuum. The total CPT charge of the universe is zero.

Boyle, Finn, and Turok showed that imposing CPT symmetry as a cosmological boundary condition has concrete observable consequences:

1. Dark matter candidate: The model has three right-handed neutrinos. A Z_2 symmetry makes the lightest one stable, and the CPT-symmetric vacuum, which differs from the late-time vacuum, produces it at the Bang rather than thermally. It matches the observed dark matter density for a mass of about 4.8×10^8 GeV.

1. Matter excess: The two heavier right-handed neutrinos are unstable and generate the matter excess in our half of the universe through standard thermal leptogenesis; the mirror half carries the opposite excess.

1. No inflation needed: They argue that CPT symmetry, rather than inflation, accounts for the flat, radiation-dominated initial state and for the phases of the CMB acoustic oscillations, and they predict that there are no primordial long-wavelength gravitational waves.

1. Prediction for neutrino masses: The model predicts that the lightest neutrino is exactly massless.

The Boyle–Finn–Turok model shares one principle with our hypothesis, CPT symmetry as a principle of cosmology, but nothing else. Boyle, Finn, and Turok achieve CPT symmetry through a boundary condition on standard fields, with normal gravity for antimatter, while our hypothesis tries to achieve it through negative gravitational mass separating the matter and antimatter sectors.

One could try to combine the two by supposing that gravitational repulsion between matter and antimatter supplies a dynamical reason for the CPT-symmetric boundary condition, with antimatter pushed toward the $t < 0$ sector. This is Conjecture 3 again, and it fails for the same reasons: repulsion acts in space, not in time, and Claim 1 is ruled out for antihydrogen by ALPHA-g (§5.1). The Boyle–Finn–Turok model does not need it.

3.4 Formal Development: Gravitational CPT and Temporal Segregation

We now set out the temporal segregation conjecture more formally and show where it fails to follow from the equations. Consider the Einstein field equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)\, T_{\mu\nu}$$

In a universe containing both matter (with stress-energy $T_{\mu\nu}^{(+)}$ and positive mass density $\rho > 0$) and antimatter (with stress-energy $T_{\mu\nu}^{(-)}$ and negative mass density $-\rho < 0$, per Claim 1):

$$G_{\mu\nu} = (8\pi G/c^4)(T_{\mu\nu}^{(+)} + T_{\mu\nu}^{(-)})$$

If at the initial moment $t = 0$ (the Big Bang), matter and antimatter are created in equal quantities, the total stress-energy is:

$$T_{\mu\nu}^{\text{total}} = T_{\mu\nu}^{(+)} + T_{\mu\nu}^{(-)} = 0$$

The net gravitational effect is zero, consistent with creation from "nothing" as in the Tryon (1973) and Vilenkin (1982) proposals for the universe as a vacuum fluctuation.

For a homogeneous mixture of the two components in a spatially flat universe, the Friedmann equation gives:

$$H^2 = (8\pi G/3)(\rho_+ + \rho_-) = (8\pi G/3)(\rho - \rho) = 0$$

The expansion rate vanishes. (With open spatial sections, the same cancellation instead gives the coasting Milne expansion of §7.2, Case 1, which is the route the Dirac-Milne model takes.)

Conjecture 3 gets past this result by assertion. It supposes that matter evolves toward $t > 0$ and antimatter toward $t < 0$, so that each sector separately satisfies:

$$H^2 = (8\pi G/3)\rho$$

with positive ρ in each sector's own temporal frame. Nothing in the field equations produces this split. Gravitational repulsion between matter and antimatter acts in space within a single time direction, and no dynamical equation here moves either component into a time direction. The step from a vanishing H^2 to two separately expanding sectors is postulated, not derived.

If the postulate is granted, the baryon asymmetry problem is resolved by construction: an observer in the $t > 0$ sector sees only matter and concludes that there is a matter–antimatter asymmetry, while the full CPT-symmetric universe has exact baryon symmetry. Because the conclusion is built into the assumption, it is not an independent explanation.

3.5 Relationship Between Negative Mass and Backward Time Propagation

The energy–momentum relation of special relativity:

$$E^2 = p^2c^2 + m^2c^4$$

admits both positive- and negative-energy solutions, $E = \pm\sqrt{(p^2c^2 + m^2c^4)}$. In quantum mechanics, a state of energy E evolves as:

$$\psi(t) = \psi(0)\ e^{-iEt/\hbar}$$

so a negative-energy solution evolving forward in time has the same time dependence as a positive-energy solution evolving backward in time:

$$e^{-i(-|E|)t/\hbar} = e^{+i|E|t/\hbar} = e^{-i|E|(-t)/\hbar}$$

This identity is the mathematical root of the Feynman–Stückelberg interpretation, but it points the opposite way from Conjecture 3. Feynman and Stückelberg used it to remove negative energies from physics: the negative-energy solution running backward in time is reinterpreted as a positive-energy antiparticle running forward in time (§2.1). Every antiparticle that is actually produced and detected carries positive energy, as the 1.022 MeV released in electron–positron annihilation shows (§8.4). In standard general relativity, where gravity couples to energy and momentum, the interpretation therefore predicts that antimatter gravitates normally, the same conclusion §2.5 reached from the modern QFT picture.

Identifying negative energy with negative mass through $E = mc^2$ does not rescue the conjecture. The negative-energy solutions have the same positive mass m (only m^2 enters the energy–momentum relation), and after the reinterpretation the physical antiparticle has positive energy as well. The tempting equivalence "negative mass forward in time is positive mass backward in time" therefore does not hold for real antiparticles. The Feynman–Stückelberg interpretation, correctly applied, provides no link between antimatter and negative gravitational mass, and Conjecture 3 has to stand, if at all, as an independent assumption.

SECTION IV

4. Cosmological Implications

4.1 Big Bang Nucleosynthesis

Big Bang nucleosynthesis (BBN) is one of the most precisely tested pillars of standard cosmology. The predicted abundances of light elements (^{4}He, D, ^{3}He, ^{7}Li) depend sensitively on the baryon-to-photon ratio η_B, the number of relativistic species N_{eff}, and the neutron lifetime τ_n.

In the negative-mass antimatter framework, the key question is whether the $t > 0$ matter sector undergoes BBN identically to the standard model. Under Conjecture 3, if all antimatter has been assigned to the $t < 0$ sector by the time of nucleosynthesis ($T \sim 1$ MeV, $t \sim 1$ s), then BBN in the matter sector proceeds exactly as in standard cosmology, with no modifications.

The critical test is the timescale of matter–antimatter separation relative to the timescale of BBN. If gravitational repulsion is the only mechanism for separation, we need to estimate whether separation is complete by $t \sim 1$ s.

In the Dirac-Milne cosmology of Benoit-Lévy and Chardin (2012), matter and antimatter coexist in the same temporal sector but are gravitationally segregated into spatial domains, and annihilation continues at the domain boundaries. Nucleosynthesis in that model faces serious problems. In its coasting expansion, the observed helium-4 abundance requires a baryon-to-photon ratio between about 8×10^{-9} and 9.6×10^{-9}, some 13 to 16 times the CMB value (Benoit-Lévy & Chardin, 2009, 2012). Thermal nucleosynthesis at that ratio destroys almost all primordial deuterium and helium-3, so the model relies on deuterium produced later, by antiproton annihilation on helium-4 at the matter–antimatter boundaries, which works only for a particular domain size. Chardin and Manfredi (2018) concede that the model overproduces helium-3 by nearly a factor of 10 and that the full CMB power spectrum has yet to be established.

Under Conjecture 3, standard BBN is recovered because the conjecture removes all antimatter from our sector before nucleosynthesis. This is not an independent success: it holds by construction and would hold for any assumption that removes the antimatter.

4.2 CMB Observations

The cosmic microwave background provides the most detailed snapshot of the early universe at the time of recombination ($z \approx 1100$, $T \approx 3000$ K). The CMB power spectrum is sensitive to:

- The baryon density $\Omega_b h^2$ (through odd/even peak ratios)
- The matter density $\Omega_m h^2$ (through the matter-radiation equality epoch)
- The spatial curvature Ω_k (through the angular diameter distance to last scattering)
- The amplitude and spectral index of primordial perturbations (A_s, n_s)

In the Boyle–Finn–Turok CPT-symmetric model, the CMB predictions are modified primarily through:

1. The absence of inflation, which changes the mechanism for generating primordial perturbations
1. The presence of a specific dark matter candidate (a right-handed neutrino) with specific properties
1. The prediction that the lightest neutrino is massless
1. The prediction that there are no primordial long-wavelength gravitational waves, and hence no primordial B-mode polarization

Boyle and Turok (2021) later proposed that dimension-zero scalar fields, introduced to cancel the vacuum energy and Weyl anomaly of the Standard Model, would carry a scale-invariant power spectrum extending to long wavelengths, which could explain the primordial perturbations without inflation. The proposal has not yet been confronted with a fit to CMB data.

If, under Conjecture 3, negative-mass antimatter supplied a dynamical mechanism for CPT symmetry, additional observational signatures might include:

- Modified primordial gravitational wave spectrum (since gravitational waves probe the full matter content, including any residual negative-mass component)
- Specific signatures in CMB polarization (the B-mode spectrum)
- Modified Sachs-Wolfe effect at large angular scales

4.3 Dark Energy and the Cosmological Constant

Perhaps the most intriguing cosmological implication of the negative-mass antimatter hypothesis concerns dark energy. The observed accelerated expansion of the universe, discovered by Riess et al. (1998) and Perlmutter et al. (1999), is parameterized by a cosmological constant Λ or equivalently a dark energy density:

$$\rho_\Lambda = \Lambda c^2 / (8\pi G) \approx 5.8 \times 10^{-27}\ \text{kg/m}^3$$

corresponding to $\Omega_\Lambda \approx 0.685$ for a Hubble constant of 67.4 km/s/Mpc (Planck Collaboration, 2020).

The cosmological constant problem (why ρ_Λ is 120 orders of magnitude smaller than naive quantum field theory estimates) is arguably the worst prediction in all of physics.

Negative mass has been proposed as a route to dark energy (Farnes, 2018). Under Conjecture 3, one might suppose that the negative-mass antimatter of the $t < 0$ sector exerts a residual gravitational influence on the $t > 0$ sector through the boundary conditions at $t = 0$. No mechanism for such an influence has been worked out, and the direct cosmological effect of a negative mass density has the wrong sign, as Farnes' model shows.

Farnes (2018) built a concrete negative-mass cosmology. He added a creation term to the field equations, analogous to the creation field of steady-state cosmology (Hoyle & Narlikar, 1964), so that negative mass is continuously created and its density stays constant as the universe expands. The modified Friedmann equation is:

$$H^2 = (8\pi G/3)(\rho_+ + \rho_-)$$

with $\rho_- < 0$ held constant. Farnes identifies the constant term with a cosmological constant:

$$\Lambda = 8\pi G\rho_- / c^2 < 0$$

and notes that, because ρ_- is negative, so is Λ. A constant negative density lowers H^2, so it acts as a negative cosmological constant, which slows the expansion rather than accelerating it (§7.2, Case 3). Farnes' own solution is a cyclic universe that expands to a maximum and recollapses, and he describes the acceleration in his model as local rather than global.

Farnes did not fit his model to supernova, BAO, or CMB data. His abstract claims only that the model may have the potential to be consistent with such observations, and he argues that the supernova data may not require a positive Λ. His one CMB estimate, which borrows the Dirac-Milne calculation of

Benoit-Lévy and Chardin (2012), puts the multipole of the acoustic scale near 160, against about 300 measured. The model is a toy model that illustrates some qualitative features of negative mass; it does not show that negative mass can mimic dark energy.

4.4 Large-Scale Structure

The formation of large-scale structure in the universe depends on the growth of gravitational instabilities from primordial density perturbations. In the standard ΛCDM model, cold dark matter provides the gravitational scaffolding for structure formation, with baryons falling into dark matter potential wells after decoupling from photons at recombination.

In the negative-mass antimatter framework, structure formation is modified in several ways:

1. Modified growth rate: If any negative-mass component remains in our sector, it alters the growth of structure, which must stay consistent with observations of the matter power spectrum $P(k)$ and the growth rate $f\sigma_8$.

1. Void structure: Antimatter obeying Claim 1, with positive inertial mass, is repelled by matter and would collect in cosmic voids, where it would push matter outward toward walls and filaments. If any such residual negative mass exists in the t > 0 sector, voids would be emptier of matter than in ΛCDM. This could bear on the "void phenomenon," Peebles' argument that voids contain fewer galaxies than cold dark matter models predict (Peebles, 2001).

1. Galaxy rotation curves: Farnes' negative masses are a different kind of matter. They obey the equivalence principle, with negative inertial and negative gravitational mass, so they are attracted toward positive-mass galaxies while repelling one another. In his N-body simulations they fall toward a galaxy, oscillate through it, and settle into a halo several galactic radii across, which flattens the rotation curve qualitatively (Farnes, 2018). Antimatter obeying Claim 1 would instead be repelled by galaxies, so these halos cannot be made of Claim 1 antimatter. The simulations were

toy models, with direct-summation codes of up to 50,000 particles and no matter creation. Socas-Navarro (2019) found that the predicted halos do not match the observed shapes and densities of galactic dark matter halos, that stability would require them to be less massive than the baryonic component, and that the model suffers a large-scale runaway effect in which galaxies would move in random directions at nearly the speed of light.

SECTION V

5. Experimental Evidence and Constraints

5.1 The ALPHA-g Experiment at CERN

The most significant experimental development for our hypothesis came in September 2023, when the ALPHA-g collaboration at CERN published the first direct measurement of the gravitational behavior of antimatter (Anderson et al., 2023, Nature 621, 716–722).

ALPHA-g holds antihydrogen atoms (each a bound state of an antiproton and a positron, electrically neutral) in a vertical magnetic trap. Slowly lowering the magnetic barriers at the top and bottom of the trap lets the atoms escape either upward or downward, and detectors record where each one annihilates on the trap wall. Gravity tips the balance: in a vertically symmetric trap, an atom that falls normally is more likely to leave through the bottom, and simulations of hydrogen under ALPHA-g conditions put about 80% of the escapes there. By repeating the release with the barriers deliberately unbalanced by known amounts, the collaboration fitted the gravitational acceleration directly.

The result: antihydrogen falls down. The measured local acceleration of antihydrogen toward Earth is:

$$\bar{g} = (0.75 \pm 0.13\ (\text{statistical + systematic}) \pm 0.16\ (\text{simulation}))\ g$$

where $g = 9.81\ \text{m/s}^2$ is the acceleration of ordinary matter. The result is consistent with $\bar{g}/g = 1$ (normal gravity). The collaboration reports that the probability that its data are consistent with repulsive gravity, $\bar{g}/g = -1$, is less than 10^{-15}.

This result rules out the simplest version of our hypothesis (Claim 1 as stated). If antihydrogen had negative passive gravitational mass, it would fall upward; ALPHA-g shows that it falls downward.

5.2 Interpretation of ALPHA-g Results

Several caveats affect how far the result reaches, though none of them restores the hypothesis:

1. Precision: The ALPHA-g result is consistent with normal gravity but has uncertainties of about 20% of g. It does not yet test the weak equivalence principle at high precision. There is room for a small anomalous gravitational component, but not for the full reversal that Claim 1 requires.

2. What is measured: ALPHA-g measures the gravitational behavior of antihydrogen (one antiproton + one positron) in Earth's gravitational field. This is the gravitational interaction of antimatter with the overwhelmingly matter-dominated local environment. If our hypothesis involves a more subtle modification (for example, if the gravitational mass sign depends on the cosmological sector rather than the particle identity), then ALPHA-g might not be testing the relevant regime. Any such version, however, gives up the property that was supposed to drive the cosmology: if antimatter in our sector gravitates normally, there is no repulsion to separate matter from antimatter.

3. Composition: An antiproton's mass is mostly the energy of the quark and gluon fields that bind it. The rest masses of its three valence antiquarks supply only about 1% of it: the up and down quark masses are about 2.2 and 4.7 MeV, against 938 MeV for the proton (Navas et al., 2024). If that field

energy gravitates normally, as §8.6 argues for self-conjugate fields such as the gluon field, antihydrogen would fall down at nearly the normal rate even if Claim 1 held for the elementary antiquarks and the positron (the constituent rule of §8.6). ALPHA-g therefore tests Claim 1 in the form in which the sign of gravitational mass belongs to the antiproton as a whole. Hajdukovic (2011) explored a related idea, that particles and antiparticles carry gravitational charges of opposite sign, so that virtual pairs act as gravitational dipoles and polarize the quantum vacuum in a way that might mimic dark matter. His proposal concerns the vacuum, not bound antimatter, and rests on the same premise that ALPHA-g has now contradicted for antihydrogen.

4. The temporal-segregation version: The version built on Conjecture 3, in which antimatter ended up in the $t < 0$ sector at the Big Bang, is not directly tested by ALPHA-g. ALPHA-g tests how antihydrogen (produced and confined in the $t > 0$ sector) behaves in a local gravitational field. If temporal segregation were a cosmological boundary effect rather than a local force, antihydrogen created in the laboratory might behave normally in local gravity while the cosmological mechanism operated through different physics. But this concedes Claim 1, the only proposed driver of segregation, so this version is left with no mechanism at all.

5.3 AEGIS and GBAR Experiments

Two other antimatter gravity experiments at CERN are pursuing complementary approaches:

AEgIS (Antimatter Experiment: Gravity, Interferometry, Spectroscopy) aims to measure the gravitational acceleration of antihydrogen by passing a horizontal antihydrogen beam through a Moiré deflectometer, targeting a precision of 1% on $\bar{g}/g$ in its first phase (Kellerbauer et al., 2008).

GBAR (Gravitational Behaviour of Antihydrogen at Rest) takes a different approach, producing ultracold antihydrogen ions ($\overline{H}^+$), cooling them sympathetically, then photodetaching the extra positron and observing the free-fall trajectory of the resulting neutral $\overline{H}$ (Pérez et al., 2015). GBAR aims for precision at the percent level and eventually much better.

As of 2025, neither AEgIS nor GBAR had reported a measurement of the gravitational acceleration of antihydrogen. AEgIS considered a first result from pulsed, horizontally travelling antihydrogen feasible before CERN's third long shutdown, with more sensitive measurements after it, and ALPHA-g had reproduced its 2023 result and expected to reach a precision of 1% or better with its existing apparatus (Caravita et al., 2025).

These experiments will progressively tighten the constraints on anomalous gravitational behavior of antimatter. The naive negative-mass hypothesis is already excluded by ALPHA-g; the new experiments will test much smaller departures from $\bar{g}/g = 1$. A small deviation (for example, $\bar{g}/g = 1.001$ or 0.999) would be a discovery of enormous significance, though on its own it would not restore the gravitational repulsion this framework needs.

5.4 Indirect Constraints

Several indirect constraints on the gravitational properties of antimatter exist:

1. Positronium free-fall: Positronium (an electron–positron bound state) contains equal parts matter and antimatter, and what it should do in free fall depends on a rule this framework has not fixed. If the sign of gravitational mass attaches to each constituent, positronium's gravitational mass nearly cancels and it would barely fall. If, as §8.6 argues for self-conjugate systems, a system that is its own antiparticle gravitates by its total energy, positronium falls normally. A positronium free-fall measurement would therefore decide between these two rules rather than test the hypothesis as a whole. Direct positronium free fall has not been measured.

2. Kaon oscillations and CP violation: Chardin and Rax (1992) argued that the observed CP violation in the neutral kaon system could be explained by a gravitational energy difference between the K^0 (a d quark and an $\bar{s}$ antiquark) and the $\overline{K}^0$ (an s quark and a $\bar{d}$ antiquark) in Earth's gravitational field. However, this explanation has been challenged by the observation that CP violation in the B-meson system has a different pattern than the gravitational explanation would predict.

3. Equivalence principle tests and virtual antimatter: Atoms contain antimatter in virtual and sea form. Alves, Jankowiak, and Saraswat (2009) estimate that sea antiquarks carry about 10% of a nucleon's energy, and that because this content differs slightly from element to element, the equivalence-principle tests then available bounded any difference between the free-fall accelerations of antihydrogen and hydrogen at about one part in 10^9. The bound assumes that virtual and sea antimatter gravitates the way free antimatter would. MICROSCOPE's later results (Touboul et al., 2017, 2022) tighten the underlying equivalence-principle limits.

This is perhaps the strongest indirect argument against our hypothesis: if virtual antimatter within matter had negative gravitational mass, the gravitational mass of different elements (which have different virtual antimatter content) would differ, violating the equivalence principle at a level detectable by existing experiments. However, this argument assumes that virtual pairs contribute to gravitational mass in the same way as real particles — an assumption that is plausible but not proven.

5.5 Neutrino-Antineutrino Oscillations

If neutrinos and antineutrinos have different gravitational masses, their oscillation behavior in gravitational fields would differ. The MINOS experiment and subsequent long-baseline neutrino experiments have measured neutrino and antineutrino oscillation parameters. The CP-violating phase δ_{CP} in neutrino oscillations, which T2K has begun to constrain (T2K Collaboration, Abe et al., 2020), could potentially receive contributions from gravitational effects if our hypothesis is correct. However, the current precision of neutrino oscillation measurements is insufficient to distinguish gravitational CP violation from intrinsic CP violation.

6. Relationship to Dark Matter and Dark Energy

6.1 The Dark Sector Problem

The standard ΛCDM cosmological model requires two components with no known particle physics counterpart:

- Dark matter ($\Omega_{DM} \approx 0.265$): gravitationally attractive, non-baryonic, cold, collisionless
- Dark energy ($\Omega_{\Lambda} \approx 0.685$): gravitationally repulsive, uniform, constant density

Together, these "dark" components constitute ~95% of the energy density of the universe, while the ordinary matter we understand constitutes only ~5%. This situation has been described as a crisis for fundamental physics.

6.2 Farnes' Negative-Mass Cosmology

Farnes (2018) proposed a toy model in which a single component, a negative-mass fluid, accounts for both dark matter and dark energy. The key features of the model are:

1. Continuous creation: Negative-mass particles are continuously created, keeping the negative-mass density ρ_- constant as the universe expands, by analogy with the C-field of Hoyle and Narlikar (1964) in steady-state cosmology.

1. Cosmological constant term: The constant negative-mass density is identified with a cosmological constant, $\Lambda = 8\pi G\rho_-/c^2$. Because $\rho_- < 0$, this Λ is negative (§4.3, §7.2). The model is cyclic: the universe expands, reaches a maximum, and recollapses. Rather than deriving the observed acceleration, Farnes argues that the supernova data may not require a positive Λ.

1. Halo formation: Farnes' negative masses obey the equivalence principle, so they are attracted toward positive-mass galaxies while repelling one another. In his N-body simulations they settle into halos several galactic radii across, which flatten galaxy rotation curves qualitatively.

1. Structure formation: In a cosmological-scale toy simulation, positive and negative masses together formed filaments, voids, and clusters, and Farnes suggests that the model's time-variable Hubble parameter could bear on current tensions between cosmological measurements.

Socas-Navarro (2019) examined the model critically and found several problems: the predicted halos do not match the observed shapes and densities of galactic dark matter halos; the halos would have to be less massive than the baryonic component to remain stable; and, most seriously, a large-scale version of the runaway effect would send galaxies moving in random directions at nearly the speed of light.

6.3 Connecting Farnes' Model to the Antimatter Hypothesis

The hope behind connecting the two was a framework in which most antimatter sits in the $t < 0$ sector (Conjecture 3), residual antimatter in our sector supplies dark energy and dark matter phenomenology, and Farnes' continuous creation is reinterpreted as pair production from the vacuum.

It does not work. Farnes' negative-mass fluid cannot be identified with antimatter as Claim 1 describes it. Farnes' negative masses have negative inertial mass as well as negative gravitational mass, so they obey the equivalence principle and fall toward positive masses, which is what lets them gather into halos around galaxies. Claim 1 antimatter has positive inertial mass and negative gravitational mass, so it would be repelled by galaxies and could not form such halos. The two are different kinds of matter with opposite dynamics. Several further problems would remain even if the identification were attempted:

1. Inertial mass: Farnes' model needs negative inertial mass, but CPT requires antiparticles to have the same positive inertial mass as particles.

1. ALPHA-g: Antihydrogen falls down, which Claim 1 forbids.

1. Pair production: Pair production from the vacuum creates matter and antimatter in equal amounts, so it cannot supply a net negative-mass density.

1. Annihilation: Why doesn't the negative-mass antimatter in the halo annihilate with the positive-mass matter in the galaxy? (Farnes' model uses a generic negative-mass fluid, not specifically antimatter, partly to avoid this issue.)

1. BBN constraints: Significant quantities of free antimatter at T ~ 1 MeV would modify nucleosynthesis through annihilation processes.

1. Halo problems: The objections of Socas-Navarro (2019) apply to the halo picture whatever the halo is made of.

6.4 The Boyle–Finn–Turok Dark Matter Candidate

The CPT-symmetric universe model of Boyle, Finn, and Turok offers a different connection to dark matter, and it involves no negative mass. Their model contains three right-handed neutrinos (as in the seesaw mechanism). A Z_2 symmetry makes the lightest of them stable, and because the CPT-symmetric vacuum differs from the late-time vacuum, this neutrino is produced at the Bang rather than thermally.

Boyle, Finn, and Turok (2018) found that this stable right-handed neutrino has the observed dark matter abundance if its mass is approximately:

$$m_{\mathrm{N}} \approx 4.8 \times 10^{8}\ \mathrm{GeV}$$

This is a remarkably specific prediction from a minimal framework. The dark matter candidate is "ordinary" in the sense that it has positive mass and does not require exotic physics, yet its abundance follows from the CPT-symmetric boundary condition.

Combining the Boyle–Finn–Turok framework with our negative-mass antimatter hypothesis would give a picture where:

- Dark matter is the stable right-handed neutrino of the CPT-symmetric model
- Dark energy comes from some residual gravitational effect of the $t < 0$ antimatter sector (Conjecture 3)
- The baryon asymmetry is handled by temporal segregation (Conjecture 3)

The combination adds nothing the Boyle–Finn–Turok model needs. That model already accounts for the matter excess in our half through thermal leptogenesis driven by the two heavier right-handed neutrinos, with the opposite excess in the mirror half, and it uses no negative mass. The dark-energy and segregation elements rest on Conjecture 3 and have no mechanism behind them.

6.5 Alternative: Negative-Mass Antimatter as Dark Energy Only

A more conservative version of the hypothesis abandons the attempt to explain dark matter through negative mass and focuses solely on dark energy. In this version:

- Dark matter is a separate, conventional (positive-mass) particle (e.g., a WIMP or axion)
- Antimatter has negative gravitational mass and has been temporally segregated
- The boundary between the $t > 0$ and $t < 0$ sectors imposes effective boundary conditions on the gravitational field that produce accelerated expansion

In this version the accelerated expansion of the universe would be a consequence of the universe's CPT-symmetric structure rather than requiring a separate dark energy component. It is less ambitious, but it inherits the same unsupported premises: it depends on Claim 1, which ALPHA-g excludes for antihydrogen, and on Conjecture 3, and no boundary condition that produces accelerated expansion has been written down.

SECTION VII

7. Mathematical Formalism

7.1 Modified Einstein Field Equations with Negative Mass

We begin by writing the Einstein field equations for a universe containing both positive-mass matter (density ρ_+, pressure p_+) and negative-mass antimatter (density $\rho_- < 0$, pressure p_-):

$$G_{\mu\nu} = (8\pi G/c^4)(T_{\mu\nu}^{(+)} + T_{\mu\nu}^{(-)})$$

For a perfect fluid with negative mass:

$$T_{\mu\nu}^{(-)} = (\rho_- + p_-/c^2)\, u_\mu u_\nu + p_- g_{\mu\nu}$$

where $\rho_- < 0$ and we take $p_- = w_- \rho_- c^2$ with equation of state parameter w_-.

7.2 Modified Friedmann Equations

For a homogeneous, isotropic universe (FLRW metric) containing both positive- and negative-mass components:

$$ds^2 = -c^2dt^2 + a(t)^2\,[dr^2/(1 - kr^2) + r^2d\Omega^2]$$

The Friedmann equations become:

$$H^2 \equiv (\dot{a}/a)^2 = (8\pi G/3)(\rho_+ + \rho_-) - kc^2/a^2$$

$$\ddot{a}/a = -(4\pi G/3)[(\rho_+ + \rho_-) + 3(p_+ + p_-)/c^2]$$

Case 1: Equal and opposite matter and antimatter ($\rho_+ + \rho_- = 0$).

If the universe contains equal quantities of positive and negative mass:

$$H^2 = -kc^2/a^2$$

For $k = -1$ (open universe), $H^2 = c^2/a^2$, giving $a(t) = ct$. This is the coasting or Milne cosmology, which has $\ddot{a} = 0$ and is neither accelerating nor decelerating. This is the basis of the Dirac-Milne cosmology (Benoit-Lévy & Chardin, 2012).

Case 2: Temporal segregation with residual negative mass (under Conjecture 3).

If most antimatter is in the t < 0 sector but a fraction f remains in t > 0:

$$\rho_{total} = \rho_+ - f\rho_+ = (1 - f)\rho_+$$

The Friedmann equation for the matter-dominated era becomes:

$$H^2 = (8\pi G/3)(1 - f)\rho_+ - kc^2/a^2$$

For the acceleration equation, with pressureless matter ($p_+ = 0$) and $p_- = w_-\rho_-c^2$:

$$\ddot{a}/a = -(4\pi G/3)(1 - f)\rho_+ + (4\pi G/3) \cdot 3fw_-\rho_+ = -(4\pi G/3)\rho_+[1 - f(1 + 3w_-)]$$

The negative-mass component's own contribution, $-(4\pi G/3)\rho_-(1 + 3w_-)$, is positive, and so pushes toward acceleration, only if $w_- > -1/3$. A negative density reverses the familiar rule, under which a positive-density component accelerates the expansion only if $w < -1/3$. The total expansion accelerates only if $f(1 + 3w_-) > 1$. For dust-like negative mass ($w_- = 0$) that requires f > 1, a net negative density, which is incompatible with $H^2 > 0$ in a spatially flat universe.

Case 3: Farnes' continuously created negative mass.

Farnes (2018) adds a creation term so that ρ_- stays constant as the universe expands, with new negative-mass material continuously created. The continuity equation with a source term is:

$$\dot{\rho_-} + 3H(\rho_- + p_-/c^2) = \Gamma$$

where Γ is the rate at which mass density is created. Requiring $\dot{\rho_-} = 0$ gives:

$$\Gamma = 3H(\rho_- + p_-/c^2)$$

For $\rho_- < 0$ and dust-like negative matter ($p_- = 0$):

$$\Gamma = 3H\rho_- = -3H|\rho_-| < 0$$

The creation rate is negative because what is created is negative mass: the mass density added at each instant is below zero. The Friedmann equation becomes:

$$H^2 = (8\pi G/3)\rho_+ + (8\pi G/3)\rho_- = (8\pi G/3)\rho_+ - (8\pi G/3)|\rho_-|$$

The constant negative-mass density enters exactly as a negative cosmological constant, the identification Farnes himself makes:

$$\Lambda_{\text{eff}} = 8\pi G\rho_-/c^2 = -8\pi G|\rho_-|/c^2 < 0$$

Differentiating the Friedmann equation with $\dot{\rho}_\Lambda = 0$ shows that the constant term contributes $(8\pi G/3)\rho_\Lambda < 0$ to $\ddot{a}/a$, the same as a fluid with $w = -1$, so it decelerates the expansion. A negative cosmological constant cannot produce the observed acceleration; it drives the expansion to a maximum and then to recollapse, which is the cyclic behavior of Farnes' solution.

7.3 Negative Mass in the Schwarzschild Solution

The Schwarzschild solution for a point mass M is:

$$ds^2 = -(1 - 2GM/c^2 r)c^2 dt^2 + (1 - 2GM/c^2 r)^{-1} dr^2 + r^2 d\Omega^2$$

For M < 0 (negative mass):

$$ds^2 = -(1 + 2G|M|/c^2 r)c^2 dt^2 + (1 + 2G|M|/c^2 r)^{-1} dr^2 + r^2 d\Omega^2$$

Key features of the negative-mass Schwarzschild solution (Bonnor, 1989):

1. No horizon: The factor $(1 + 2G|M|/c^2 r) > 1$ for all $r > 0$, so there is no event horizon. The singularity at $r = 0$ is naked.

1. Repulsive gravity: Test particles are repelled. The geodesic equation in the Newtonian limit gives:

$$\mathrm{d}^2 r/\mathrm{d}t^2 = +G|M|/r^2$$

1. Negative Komar mass: The Komar mass integral yields $M_K = M < 0$, consistent with the ADM mass.

1. Gravitational lensing: A negative-mass object acts as a *diverging* gravitational lens, like a concave lens, spreading light rays apart rather than focusing them. It would distort background images radially rather than tangentially, a signature that lensing surveys could in principle detect (Izumi et al., 2013).

7.4 Geodesic Equation for Negative-Mass Test Particles

The geodesic equation for a test particle is:

$$d^2x^\mu/d\tau^2 + \Gamma^\mu{}_{\alpha\beta}\,(dx^\alpha/d\tau)(dx^\beta/d\tau) = 0$$

This is independent of the mass of the test particle (this is the universality of free fall, the equivalence principle). If we maintain the equivalence principle but allow both m_i and the gravitational mass to be negative (Bondi's approach), then negative-mass test particles follow the same geodesics as positive-mass particles; they fall toward positive masses just as positive masses do.

If instead the equivalence principle is violated, with passive gravitational mass $m_p < 0$ while $m_i > 0$ as Claim 1 requires, the particle cannot follow the geodesics of ordinary matter. In the Newtonian limit its equation of motion is:

$$m_i\, d^2\mathbf{r}/dt^2 = -m_p \nabla\Phi$$

$$d^2\mathbf{r}/dt^2 = -(m_p/m_i)\nabla\Phi$$

For $m_p/m_i = -1$ (our antimatter hypothesis), this gives $d^2\mathbf{r}/dt^2 = +\nabla\Phi$: the acceleration is reversed, and the antiparticle falls up in a gravitational field.

This violation of the equivalence principle is the most radical aspect of our hypothesis and the one most directly constrained by experiment.

7.5 Quantum Field Theory of Gravitationally Repulsive Antimatter

A proper quantum field theory treatment requires coupling the Dirac field to gravity. In curved spacetime, the Dirac equation is:

$$(i\gamma^\mu\nabla_\mu - mc/\hbar)\psi = 0$$

where $\nabla_\mu = \partial_\mu + (1/4)\omega_\mu{}^{ab}\gamma_a\gamma_b$ is the spinor covariant derivative and $\omega_\mu{}^{ab}$ is the spin connection.

Under CPT, the Dirac field transforms, up to a phase and the choice of γ-matrix representation, as:

$$\psi(x) \to \eta_{CPT}\,\gamma^5\,\psi^*(-x)$$

where η_{CPT} is a phase. One might look for antigravity in a sign change of the Dirac mass term, writing the antimatter field as obeying:

$$(i\gamma^\mu \nabla_\mu + mc/\hbar)\psi_{CPT} = 0$$

This is not Villata's argument, which is classical and concerns the geodesic and field equations (§3.2). If the gravitational coupling depended on the sign of the mass term, the CPT-transformed field (antimatter) would couple with opposite sign to gravity.

However, the sign of the Dirac mass is not physical. The field redefinition $\psi \to \gamma^5\psi$ flips it while leaving the kinetic term unchanged, and because γ^5 anticommutes with every γ_a, it commutes with the spin-connection term in ∇_μ, so the redefinition works equally well in curved spacetime. With the standard minimal coupling to gravity, a sign change of the mass term therefore has no gravitational consequence. Obtaining one requires a nonstandard coupling between spinors and gravity, which this framework does not supply.

7.6 Energy Conditions and Negative Mass

General relativity constrains the stress-energy tensor through various energy conditions:

1. Weak Energy Condition (WEC): $T_{\mu\nu}u^\mu u^\nu \geq 0$ for all timelike u^μ. This implies $\rho \geq 0$.

1. Strong Energy Condition (SEC): $(T_{\mu\nu} - \tfrac{1}{2}Tg_{\mu\nu})u^\mu u^\nu \geq 0$. This implies $\rho + 3p/c^2 \geq 0$.

1. Null Energy Condition (NEC): $T_{\mu\nu}k^\mu k^\nu \geq 0$ for all null k^μ.

1. Dominant Energy Condition (DEC): $T_{\mu\nu}u^\mu$ is non-spacelike for timelike u^μ.

Negative-mass matter violates the WEC ($\rho < 0$). Notably, the cosmological constant (dark energy) already violates the SEC: $\rho_\Lambda + 3(-\rho_\Lambda) = -2\rho_\Lambda < 0$. The energy conditions are not fundamental laws;

they are conjectures about the behavior of classical matter. Quantum effects routinely violate the WEC (for example, the Casimir effect produces regions of negative energy density).

If negative-mass antimatter exists, the WEC is violated, but this is not necessarily fatal to the theory. The key requirement is that the total energy of any isolated system remains non-negative to ensure stability of the vacuum. Under Conjecture 3, the total gravitational mass-energy of the universe, summed over both temporal sectors, would be exactly zero, consistent with creation from nothing; but this rests on the conjecture, and Claim 1 still leaves positive inertial energy in each sector (§8.4).

SECTION VIII

8. Criticisms and Counterarguments

8.1 The Runaway Paradox

The objection: Bondi (1957) showed that a positive-mass particle and a negative-mass particle, interacting gravitationally, form a "runaway pair" — the negative mass chases the positive mass, both accelerating without limit. This appears to violate energy conservation and common sense.

Response: The runaway pair actually does conserve energy and momentum. The kinetic energy of the negative-mass particle is $E_k = \frac{1}{2}mv^2 < 0$ (since m < 0), and the total kinetic energy of the pair remains constant (one positive, one negative, summing to zero). Similarly, momentum is conserved because the negative-mass particle carries negative momentum. The runaway pair is counterintuitive but not inconsistent.

Under Claim 1 the runaway does not arise at all, because antimatter keeps positive inertial mass: a matter particle and an antiparticle simply repel and fly apart, conserving energy and momentum in the

ordinary way. The runaway belongs to Bondi's case of negative inertial mass, which is the case in Farnes' model. Under Conjecture 3, matter and antimatter would in any event occupy different temporal sectors and never interact locally after the Big Bang.

8.2 The ALPHA-g Result

The objection: The 2023 ALPHA-g experiment directly measured that antihydrogen falls downward in Earth's gravity, ruling out antigravity at high significance.

Response: This is the most serious experimental challenge to the hypothesis. We acknowledge that the simplest version, in which antimatter has negative gravitational mass and falls up, is ruled out by ALPHA-g.

Three responses are available, and each comes at a cost:

1. The hypothesis is cosmological, not local. The temporal-segregation mechanism may operate through cosmological boundary conditions rather than through a local modification of gravity. Antihydrogen produced in the laboratory exists within the $t > 0$ sector and interacts normally with local gravity, while the cosmological CPT symmetry operates at the level of initial/boundary conditions.

1. Screening. One could posit that antimatter's gravitational mass is "screened" by the surrounding matter, so that antimatter in the matter-dominated $t > 0$ sector acquires an effective positive gravitational mass (by loose analogy with charge screening in QED). No such mechanism has been worked out.

1. The distinction may be between sectors, not particles. The Boyle–Finn–Turok framework shows that a CPT-symmetric universe can be built with no gravitational repulsion at all: individual

antiparticles in our sector behave normally, and the mirror sector is fixed by a boundary condition. Adopting it, however, means abandoning negative gravitational mass rather than rescuing it.

All three responses concede Claim 1 for the antimatter we can test, and with it the gravitational repulsion that was supposed to separate matter from antimatter. What remains is a cosmological conjecture without a mechanism; the framework loses its explanatory power and makes no specific prediction that the ALPHA-g result leaves open.

8.3 Virtual Antimatter and the Equivalence Principle

The objection: Atoms contain significant virtual antimatter (sea quarks, virtual electron-positron pairs). If virtual antimatter had negative gravitational mass, different atoms would have different ratios of gravitational to inertial mass, violating the equivalence principle. MICROSCOPE and torsion balance experiments constrain equivalence principle violations, MICROSCOPE at the 10^{-15} level (Touboul et al., 2022), seemingly ruling out any difference.

Response: This is a powerful argument in the context of local physics. Possible responses:

1. Virtual pairs may not contribute to gravitational mass in the same way as real particles. Virtual particles are off-shell ($E^2 \neq p^2c^2 + m^2c^4$) and their gravitational properties may differ. The theoretical question of how virtual particles gravitate is unsettled.

1. Renormalization. In renormalized QFT, the physical mass of a particle includes contributions from virtual pairs, but these contributions are already accounted for in the measured inertial mass. If gravitational mass equals total renormalized mass (as the equivalence principle demands), then virtual pairs do not produce a separate gravitational anomaly.

1. The distinction applies only to real, free antiparticles. Virtual antimatter within matter is part of the quantum vacuum dressing of the matter particle and gravitates normally. Only free, on-shell

antiparticles possess negative gravitational mass. This is *ad hoc* but not inconsistent.

8.4 Energy Conservation

The objection: If matter has positive energy and antimatter has negative energy, their annihilation should produce zero energy, not the observed radiation (e.g., electron-positron annihilation produces two 511 keV photons).

Response: This objection confuses gravitational mass/energy with inertial mass/energy. In our framework:

- The *inertial* mass of both particles and antiparticles is positive.
- The *gravitational* mass of antiparticles is negative.
- Annihilation converts inertial mass-energy into radiation: $e^+ + e^- \rightarrow 2\gamma$, releasing $2m_e c^2 = 1.022$ MeV. This is the inertial energy, which is positive for both.
- The *gravitational* energy balance is: the gravitational mass goes from $(+m_e) + (-m_e) = 0$ to two photons with gravitational mass $+E/c^2$ each. This appears to violate gravitational energy conservation.

This is a genuine problem. The resolution might be:

1. Photons as their own antiparticles have zero net gravitational charge in a CPT sense. The gravitational energy of annihilation photons comes from the conversion of kinetic/binding energy, not from gravitational mass.

1. The equivalence of gravitational and inertial mass for photons is well-tested (gravitational lensing, Shapiro delay), so photons definitely have positive gravitational mass. This creates a genuine

bookkeeping problem for the hypothesis.

We acknowledge this as one of the most serious theoretical difficulties with the framework.

8.5 Why Mainstream Physics Rejects Negative Mass

The mainstream rejection of negative gravitational mass for antimatter rests on several pillars:

1. The equivalence principle is the foundation of general relativity and has been tested to extraordinary precision. Violating it requires abandoning or modifying GR.

1. The CPT theorem (in its standard formulation) implies that particles and antiparticles have the same mass, lifetime, and magnitude of charge. The standard interpretation is that this includes gravitational mass. The theorem has also been extended beyond flat spacetime: Hollands (2004) proved a PCT theorem for the operator product expansion of quantum fields on general curved spacetimes. No such theorem exists for quantum gravity.

1. Positive energy theorems (Schoen & Yau, 1979; Witten, 1981) prove that the total energy of an asymptotically flat spacetime satisfying the dominant energy condition is non-negative. Negative mass violates the dominant energy condition.

1. Stability concerns: A universe containing both positive and negative mass has no ground state — energy could be extracted indefinitely by separating positive and negative mass-energy.

1. ALPHA-g: As discussed, direct measurement shows antihydrogen falls down.

These objections are serious. We note, however, that:

- The equivalence principle is empirical, not derived from first principles

- The CPT theorem has not been extended to quantum gravity

- Dark energy already violates the strong energy condition
- The stability concern would be addressed if Conjecture 3 held (total energy zero), though the conjecture itself is unmotivated
- ALPHA-g tests local gravity, not cosmological CPT, though without local repulsion the framework has no mechanism (§8.2)

8.6 Photon Self-Conjugacy Problem

The objection: The photon is its own antiparticle. If our hypothesis assigns negative gravitational mass to antiparticles, what gravitational mass does the photon have? It cannot be both positive and negative.

Response: The photon, being its own antiparticle, has zero "gravitational charge" in the matter/antimatter sense. Its gravitational interaction comes entirely from its energy-momentum, which is always positive. The mass/antimass distinction applies only to particles that are distinct from their antiparticles (Dirac-type particles). For Majorana-type particles (self-conjugate), the gravitational mass is determined by their energy content, as in standard GR.

This actually provides a consistency check: self-conjugate particles (photons, Z^0, Higgs, graviton, and potentially Majorana neutrinos) must gravitate normally. Only Dirac-type particles (quarks, charged leptons, Dirac neutrinos if applicable) have distinct antiparticles that could carry negative gravitational mass.

The rule has consequences the framework must accept. First, it extends to composite systems that are their own antiparticles, such as positronium and the neutral pion: by the same logic they gravitate by their total energy and fall normally, so positronium is not expected to float (§5.4). Second, it applies to field energy. Most of an antiproton's mass is the energy of the quark and gluon fields that bind it, and the gluon field, like the photon field, is its own charge conjugate. If that energy gravitates normally, an

antiproton's gravitational mass is close to that of a proton, and antihydrogen falls down whatever happens to the antiquarks themselves (§5.2). Keeping Claim 1 for bulk antimatter therefore needs a further rule: that the sign of gravitational mass belongs to the whole system and follows its net matter content, not its constituents. The framework does not supply such a rule, and both choices undermine it: the constituent rule makes bulk antimatter gravitate almost normally, and the whole-system rule is the version that ALPHA-g excludes.

SECTION IX

9. Conclusions and Future Directions

9.1 Summary of the Hypothesis

We have explored the hypothesis that the observed baryon asymmetry of the universe can be explained by antimatter possessing negative gravitational mass, together with the conjecture that matter and antimatter were segregated in time at the Big Bang into CPT-conjugate sectors of the universe. This hypothesis draws on:

- The Feynman–Stückelberg interpretation (antiparticles as backward-in-time particles), which, correctly applied, gives antiparticles positive energy (§3.5)
- Bondi's negative-mass framework in general relativity
- Villata's CPT argument for matter-antimatter gravitational repulsion
- Boyle, Finn, and Turok's CPT-symmetric universe model, which is a separate idea with no negative mass

- Farnes' negative-mass cosmology

9.2 Where the Hypothesis is Strong

1. Elegance and economy: The hypothesis would resolve the baryon asymmetry problem without new particles, new symmetry breaking, or new CP violation. It is not free of new physics, though: it requires antimatter to violate the equivalence principle, the foundation of general relativity, through a gravitational coupling that no current theory provides.

1. CPT symmetry as a cosmological principle: The idea that the universe as a whole respects CPT symmetry is aesthetically compelling and has been shown by Boyle, Finn, and Turok to have concrete, testable consequences (prediction of dark matter candidate, lightest neutrino mass = 0). Their model is a separate idea, however: it needs no negative mass and explains the matter excess in our half by thermal leptogenesis.

1. Creation from nothing: If Conjecture 3 held, the total gravitational mass-energy of the universe would be zero (positive mass in $t > 0$, negative mass in $t < 0$), so the universe could arise as a quantum fluctuation from nothing, connecting to the proposals of Tryon (1973) and Vilenkin (1982).

1. Falsifiability: Its most direct prediction, that antihydrogen falls up, could be tested and has been. That the idea could be checked against experiment is to its credit, even though the result went against it.

9.3 Where the Hypothesis is Weak or Speculative

1. ALPHA-g: The most direct test, whether antihydrogen falls up or down, has ruled out the simplest version of the hypothesis. Antihydrogen falls down.

1. No basis for temporal segregation: Conjecture 3 has no derivation. The Feynman–Stückelberg interpretation, on which it was meant to rest, gives real antiparticles positive energy.

1. Dark energy has the wrong sign: A constant negative-mass density acts as a negative cosmological constant, which decelerates the expansion, so the framework does not explain the observed acceleration (§4.3, §7.2).

1. Farnes' model does not fit: Farnes' negative masses obey the equivalence principle and are attracted toward galaxies, so they cannot be the antimatter of Claim 1 (§6.3).

1. Energy conservation in annihilation: The bookkeeping of gravitational energy in matter–antimatter annihilation is problematic.

1. Virtual antimatter and the equivalence principle: Indirect constraints from precision tests of the equivalence principle are severe, though not conclusive due to ambiguities in how virtual particles gravitate.

1. Composite systems: The framework has no consistent rule for the gravitational mass of composite systems and field energy (§8.6).

1. Violation of the equivalence principle: The hypothesis, in its simplest form, requires violating the foundational principle of general relativity.

1. The CPT argument is contested: The extension of CPT to the gravitational sector (necessary for Villata's argument) is not supported by a rigorous proof.

1. Quantitative predictions are lacking: The hypothesis, as developed here, makes qualitative predictions but lacks the precise quantitative predictions (e.g., CMB power spectrum) that would allow a statistical comparison with ΛCDM.

9.4 Future Directions

1. Precision antimatter gravity: ALPHA-g, AEgIS, and GBAR at CERN will progressively improve the precision of $\bar{g}/g$ measurements. The local-antigravity version of the hypothesis is already excluded; percent-level measurements would begin to test the constituent-rule variant, which predicts $\bar{g}$ slightly below g (§5.2, §8.6). Any deviation, however small, would be revolutionary.

1. CPT-symmetric cosmology: The Boyle–Finn–Turok program is developing precise predictions for the CMB and large-scale structure. Comparison with Planck and future CMB experiments (CMB-S4, LiteBIRD) can test the framework.

1. Primordial gravitational waves: The Boyle–Finn–Turok prediction that there are no primordial long-wavelength gravitational waves can be tested by searches for primordial B-mode polarization in the CMB.

1. Neutrino mass: The prediction that the lightest neutrino is massless can be tested by next-generation neutrino experiments (JUNO, DUNE, Project 8/KATRIN).

1. Gravitational wave astronomy: Negative-mass matter could in principle modify the gravitational wave background and the inspiral of compact objects containing antimatter (though natural antimatter compact objects are not expected to exist in the $t > 0$ sector), but no quantitative prediction has been made.

1. Anti-lensing searches: A negative-mass concentration would act as a diverging gravitational lens. Systematic searches for anti-lensing signatures in large surveys (Euclid, Vera Rubin Observatory) could constrain or detect negative-mass concentrations.

1. Theoretical development: A rigorous quantum gravity treatment of the gravitational properties of antimatter is needed. This requires progress on the quantization of gravity itself — arguably the biggest open problem in theoretical physics.

1. Positronium gravity: A measurement of the free-fall acceleration of positronium (a matter-antimatter bound state) would decide between the constituent rule, under which positronium's gravitational mass nearly cancels, and the self-conjugate rule of §8.6, under which it falls normally.

9.5 Final Assessment

The negative-mass antimatter hypothesis, in its simplest and most dramatic form, is now ruled out by the ALPHA-g experiment. The more sophisticated versions do not rescue the negative-mass route. If laboratory antimatter gravitates normally, as ALPHA-g shows and as §8.2 concedes, there is no gravitational repulsion to separate matter from antimatter, and the temporal segregation of Conjecture 3 was never derived from the Feynman–Stückelberg interpretation, which in fact gives antiparticles positive energy. These versions make no quantitative predictions (§9.3), so they cannot be distinguished experimentally from standard cosmology. The negative-gravitational-mass route to the baryon asymmetry is effectively excluded.

The Boyle–Finn–Turok CPT-symmetric universe is a separate idea and should be judged on its own terms. It treats CPT symmetry as a cosmological boundary condition, requires no negative mass and no locally measurable antigravity, and explains the matter excess in our half of the universe by thermal leptogenesis. It makes definite predictions (a stable right-handed neutrino of about 4.8×10^8 GeV as dark matter, an exactly massless lightest neutrino, and no primordial long-wavelength gravitational waves), and it is an active research program. The insight worth keeping from this report belongs to that model rather than to the negative-mass hypothesis: CPT symmetry may be a principle of cosmology as well as of particle physics.

The history of physics teaches that our deepest puzzles often require reconsidering our most basic assumptions. The baryon asymmetry problem may ultimately be resolved by new sources of CP violation within conventional baryogenesis, or it may point toward a radically different understanding

of the relationship between matter, antimatter, gravity, and time. The experiments and theoretical developments of the coming decade will be decisive.

Report completed February 2026. The author acknowledges that this work is speculative and is intended as a rigorous exploration of a non-standard hypothesis, not as advocacy for its correctness.

Revision note, October 2026. This revision corrects the use of the Feynman–Stückelberg interpretation, which gives antiparticles positive energy, and recasts temporal segregation as an unmotivated conjecture. It also corrects the account of the ALPHA-g measurement, the treatment of Farnes' model and the sign of a constant negative-mass density, the Dirac-Milne nucleosynthesis record, the description of the early arguments on antimatter gravity, several misattributed citations and numbers, and the equations, and it adds a verified reference list. The conclusion now states that the negative-gravitational-mass route is effectively excluded and presents the Boyle–Finn–Turok model as a separate idea. The copy archived under the paper's Zenodo DOI (10.5281/zenodo.19445820) predates this revision.

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